

1-3

A: 容器1内的气体与大气之间的压差

$$\therefore P_1 = P_A + 0.97 = 1.1 + 0.97 = \underline{2.07 \text{ bar}}$$

B: 容器1与容器2内气体的压差

$$\therefore P_2 = P_1 - P_B = 2.07 - 1.75 = \underline{0.32 \text{ bar}}$$

C: 容器2内的气体与大气之间的压差

$$\therefore P_C = 0.97 - P_2 = \underline{0.65 \text{ bar}}$$

1-5

$$\left[ (760 + 800)(133.3) + 300(9.80665) \right] / 10^5 = 2.109 \approx \underline{2.11 \text{ bar}}$$

1-7

$$P_1 V_1^{1.3} = P V^{1.3} \quad P = \frac{P_1 V_1^{1.3}}{V^{1.3}}$$

$$W = \int_{V_1}^{V_2} P dV = \int_{0.3}^{0.5} \frac{P_1 V_1^{1.3}}{V^{1.3}} dV = P_1 V_1^{1.3} \int_{0.3}^{0.5} V^{-1.3} dV$$

$$= P_1 V_1^{1.3} \left( \frac{V^{-0.3}}{-0.3} \right) \Big|_{0.3}^{0.5}$$

$$u = \frac{1}{6} \pi D^3$$

$$\frac{du}{dD} = \frac{1}{2} \pi D^2$$

$$du = \frac{1}{2} \pi D^2 dD$$

$$= \frac{600 \times 10^3 \times 0.3^{1.3}}{-0.3} (0.5^{-0.3} - 0.3^{-0.3})$$

$$= 85249.6797 \text{ J}$$

$$= \underline{85.2497 \text{ kJ}}$$

1-8

$$p = kD$$

$$150 \times 10^3 = 0.3 k \quad k = 500 \text{ mm}$$

$$W = \int_{V_1}^{V_2} P dV = \int_{D_1}^{D_2} kD d\left(\frac{1}{6} \pi D^3\right)$$

$$= \frac{1}{2} \pi \int_{D_1}^{D_2} kD^3 dD$$

$$\begin{aligned}
 &= \frac{1}{2} \pi \left( \frac{kd}{4} \right)^{0.4} \\
 &= \frac{1}{2} \pi (500000) \left( \frac{0.4^4}{4} - \frac{0.3^4}{4} \right) \\
 &= 3436.1170 \text{ J} \\
 &= \underline{\underline{3.4361 \text{ kJ}}}
 \end{aligned}$$

1-10

等温等外压  $W = p \Delta V = 0.09 \times 10^6 \times 2 = \underline{\underline{1.8 \times 10^5 \text{ J}}}$